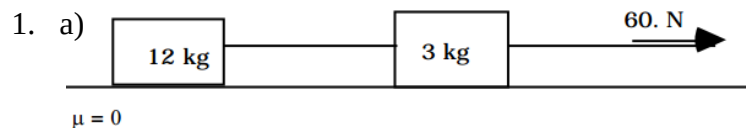


Physics 11 Practice: Forces II

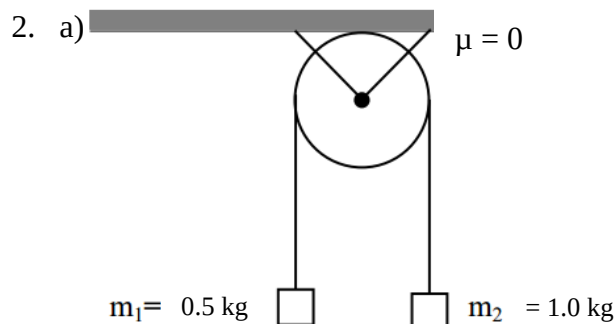
Name: _____

Date: _____ Block: _____

Find acceleration and cord tension for scenarios 1-2 (cords have negligible mass):

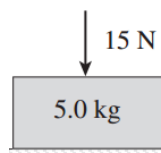


- b) Same diagram as a) but:
 $F_A = 83 \text{ N}$ [left]
 $m_{\text{left}} = 6.0 \text{ kg}$, $m_{\text{right}} = 9.5 \text{ kg}$



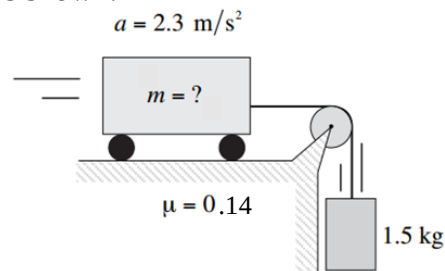
- b) Same diagram as a) but:
 $m_1 = 5.5 \text{ kg}$, $m_2 = 3.0 \text{ kg}$

3. a) A 15 N force is applied to a 5.0 kg block at rest on a level surface. Find normal force on block.



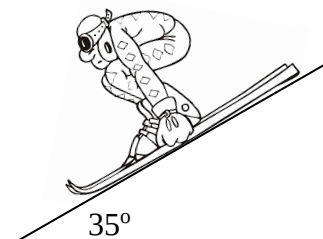
- b) 70 kg person is in an elevator accelerating downward at 1.0 m/s^2
 Find person's apparent weight (normal force)

4. a) A light cord connects a moving cart to a mass over a pulley as shown.
 Cart is accelerating at 2.3 m/s^2 . Find: i) mass of cart
 ii) cord tension



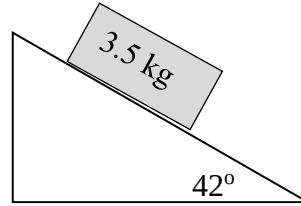
- b) Same diagram as a) but:
 $m = 4.5 \text{ kg}$, hanging mass = 3.3 kg , $\mu = 0.17$
 Find: i) acceleration of system ii) cord tension

5. A skier with a mass of 72 kg is descending a 35° slope as shown.
 Coefficient of kinetic friction between skier's skis and icy surface = 0.12 .
 Find acceleration of skier.



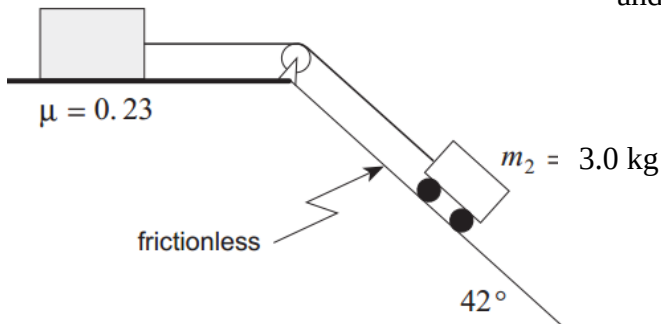
6. Block in diagram below is sliding at a steady speed down a ramp ($a = ?$)

- Find:
- normal force
 - force of friction
 - coefficient of friction



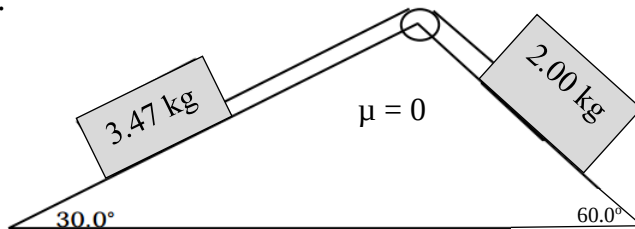
Find acceleration and cord tension for scenarios 7-9 (cords have negligible mass):

7. a) $m_1 = 5.0 \text{ kg}$

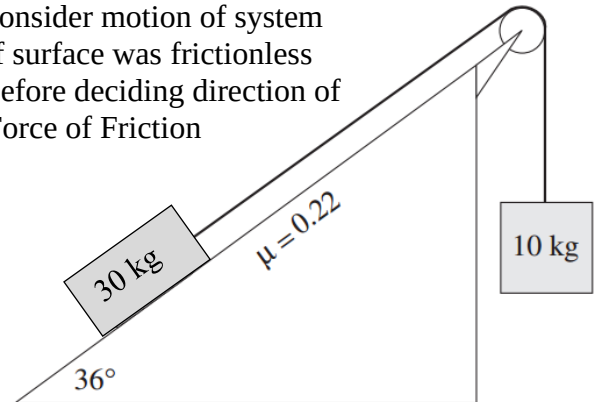


b) same diagram as a) but switch masses and use $a = 3.0 \text{ m/s}^2$ and find μ and tension

8.

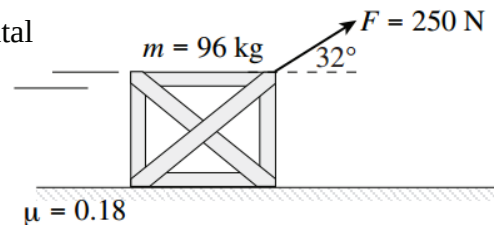


9. a) consider motion of system if surface was frictionless before deciding direction of Force of Friction



b) same diagram as a) but switch masses

10. a) 250 N pulling force applied at an angle of 32° above horizontal to a 96 kg box causes box to slide along floor as shown. Coefficient of friction between floor/box = 0.18. Find acceleration of box.



b) Same scenario as a) but with following parameters:

- $m = 68 \text{ kg}$
- $F = 193 \text{ N}$ @ 27° to horizontal pushing down/right
- $\mu = 0.11$

KEY:

1. a) 4.0 m/s^2 [right]; 48N
2. a) 3.3 m/s^2 [clockwise]; 6.5 N
3. a) 64 N
4. a) i) 3.1 kg ii) 11 N
5. 4.7 m/s^2
6. a) 25 N
7. 1.1 m/s^2 ; 17 N
8. $5.28 \times 10^{-3} \text{ m/s}^2$ toward 3.47 kg, 17.0 N
9. a) 0.56 m/s^2 toward 30 kg, 104N
10. a) 0.69 m/s^2
- b) 5.4 m/s^2 [left], 51 N
- b) 2.9 m/s^2 [counter-clockwise]; 38 N
- b) 620 N (63 kg)
- b) i) 3.2 m/s^2 [clockwise] ii) 22 N
- b) 23 N c) 0.90
- b) 0.30; 18 N
- b) 5.5 m/s^2 toward 30 kg, 130 N
- b) 1.3 m/s^2